



## Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

### 1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

|                                                                               |                                                                                                                  |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Imię i nazwisko / Name, surname                                               | Grzegorz Piotrowski                                                                                              |
| Dyscyplina kształcenia / Discipline of education                              | informatyka techniczna i telekomunikacja / information and communication technology                              |
| Wydział / Faculty                                                             | W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology                   |
| Katedra / Department                                                          | K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence                                       |
| Data rozpoczęcia kształcenia / Date of commencement of the doctoral education | 01.10.2025                                                                                                       |
| Nr ORCID                                                                      | 0009-0004-6013-0461                                                                                              |
| Promotor/ promotorzy / Supervisor(s)                                          | dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012 |
| Promotor pomocniczy / Assistant supervisor                                    | —                                                                                                                |

### 2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

Poprawa zdolności generalizacji modeli uczenia głębokiego w transferze rzeczywistość–symulacja–rzeczywistość dla fizycznej sztucznej inteligencji / Improving the generalization of deep learning models in real-to-sim-to-real transfer for physical AI

### 3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań cząstkowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:

- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,



- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

*A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:*

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

| Okres realizacji zadania (semestr kształcenia) /<br>Task implementation period (semester) | Zwięzły opis zadania / Brief description of the task                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| Semestr 1                                                                                 | <ol style="list-style-type: none"> <li>1. Literature review in the scope of sim-to-real transfer, domain randomization and domain adaptation, neural scene reconstruction from real data (such as 3D Gaussian Splatting) and deep learning for robot navigation and manipulation.</li> <li>2. Identification of the most important research gaps in the generalization of deep learning models trained in simulations built from real data.</li> <li>3. Extending theoretical knowledge, including machine learning theory and learning under distribution shift, necessary for further work on the PhD thesis.</li> <li>4. Participation in the Doctoral School courses.</li> </ol> |
| Semestr 2                                                                                 | <ol style="list-style-type: none"> <li>1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI.</li> <li>2. Initial pilot experiments on reconstructing real scenes with 3D Gaussian Splatting and importing them into navigation and manipulation simulators, with regard to the gaps identified during the literature review.</li> <li>3. Specifying the Individual Research Plan.</li> <li>4. Participation in the Doctoral School courses.</li> </ol>                                                                                                                    |



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| Semestr 3 | <ol style="list-style-type: none"><li>1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI.</li><li>2. Preparation of the research environment for robot navigation and robotic manipulation, with shared reconstruction tools: for navigation, real scene captures, digital twins in a navigation simulator and a dataset-based proxy measuring the reconstruction-fidelity gap; for manipulation, digital twins of real tabletop scenes built from images of public robot datasets and own captures, in a manipulation simulator, paired with published real-robot results of the same policies; and computing resources.</li><li>3. Conducting research on which reconstruction errors of a digital twin, varying across a scene, determine the generalization of navigation and manipulation models trained in it.</li></ol>                                                                                    |
| Semestr 4 | <ol style="list-style-type: none"><li>1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI.</li><li>2. Finishing the research on the harmful reconstruction errors in both tasks and on randomizing each region of a digital twin according to its reconstruction uncertainty and task relevance.</li><li>3. Writing a scientific article for a conference from the ministerial list (regarding the reconstruction errors that determine generalization in navigation and manipulation), planned for submission to e.g. NeurIPS 2027.</li><li>4. Starting the research on localizing the sim-to-real gap inside navigation and manipulation models, with first results on the stage at which task information is lost on real inputs.</li><li>5. Preparation of the Preludium research grant proposal to the National Science Centre.</li><li>6. Participation in an international scientific conference.</li></ol> |
| Semestr 5 | <ol style="list-style-type: none"><li>1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI.</li><li>2. Finishing the research on localizing the sim-to-real gap inside the models of both tasks and on aligning representations at that stage.</li><li>3. Writing a scientific article for a conference from the ministerial list (regarding the localization of the sim-to-real gap inside deep models), e.g. ICML 2028 or CVPR 2028.</li><li>4. Carrying out research on selecting real data under a limited budget to correct both the digital twin and the model in both tasks.</li><li>5. Starting the research on predicting whether the improvements transfer to unseen environments and between navigation and manipulation.</li></ol>                                                                                                                                                                      |



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| Semestr 6 | <ol style="list-style-type: none"> <li>1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI.</li> <li>2. Finishing the research on the selection of real data and on predicting the transfer of improvements across environments and tasks, complemented by controlled studies of physical parameters in a simulator with hidden parameters.</li> <li>3. Writing a scientific article for a conference from the ministerial list (regarding the selection of real data under a limited budget and the prediction of transfer), e.g. NeurIPS 2028.</li> <li>4. Initial validation of the developed methods on a real robot, if access to a robotics laboratory of Wrocław University of Science and Technology allows.</li> <li>5. Research internship at a foreign research group working on robot learning, if funding allows.</li> </ol> |
| Semestr 7 | <ol style="list-style-type: none"> <li>1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI.</li> <li>2. Finishing the experiments and, if possible, the validation on a real robot.</li> <li>3. Writing a scientific article for a conference or journal from the ministerial list summarizing the obtained results.</li> <li>4. Participation in an international scientific conference.</li> <li>5. Preparing the structure and sketch of the doctoral dissertation.</li> </ol>                                                                                                                                                                                                                                                                                                                                                         |
| Semestr 8 | Editing the doctoral dissertation, preparation of the final version of the document.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

#### 4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission

*Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.*

##### September 2029

*(rekomenrowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /*

*The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)*

#### 5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

*Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).*



Deep learning has brought remarkable progress in computer vision and natural language processing, largely thanks to vast amounts of data collected from the web. Physical AI, i.e., AI in robots and other embodied systems that perceive and act in the real world, cannot rely on such data. Its models need interaction data, which are slow, expensive and sometimes unsafe to collect with real robots. Therefore, such models are commonly trained in simulation. Recently, digital twins, i.e., simulations built from real data, have become a promising alternative to hand-crafted simulators: neural scene reconstruction methods, such as 3D Gaussian Splatting, turn a short recording of a real place into a photorealistic digital twin in which a model can be trained at scale. This forms a real-to-sim-to-real loop, in which real data are turned into a simulation, a model is trained there, and the model is transferred back to reality.

However, the value of this loop depends on how well the trained models generalize. A model that performs well in simulation often fails in reality, since no simulation reproduces the real world exactly: appearance, geometry, lighting and physics are only approximated, and reconstruction errors differ between regions of a scene. The resulting drop in performance, called the sim-to-real gap, is caused by the shift between the distributions of simulated and real data, which makes it a particular case of learning under distribution shift and a central obstacle to physical AI. The methods proposed so far have several limitations. Domain randomization, even when fitted to real data, varies a few global parameters regardless of where the simulation is inaccurate, and invariant representations may discard information needed for the task. Moreover, the few real data used for adaptation usually serve to correct either the model or the simulation, but not both. Finally, most methods are evaluated on a single task and a few scenes, and it is not known which properties of the digital twin and of the learned representations determine whether a model generalizes to reality.

Therefore, the dissertation aims to improve the generalization of deep learning models in real-to-sim-to-real transfer at the points of the loop where the sim-to-real gap arises. The research shall focus on understanding which reconstruction errors of a digital twin are harmful to generalization, on localizing where inside a model the gap arises and reducing it there, on choosing the small amount of real data that corrects both the digital twin and the model most efficiently, and on verifying whether the obtained improvements carry over to unseen environments and from one physical task to another, and which properties of the shift explain when they do. The methods shall not be tied to a particular task, and robot navigation and robotic manipulation shall serve as two equal testbeds. For navigation, public recordings of real scenes, in which a second, independent capture measures the reconstruction-fidelity gap, shall be complemented by a real robot, which provides the main evidence for differences in actuation and sensors. For manipulation, digital twins of real tabletop scenes shall be built from images of public robot datasets and compared with published real-robot results of the same policies. Beyond more reliable models, the research could help explain why models trained on synthetic data succeed or fail on real data, which is relevant to machine learning in general.

The results have potential applications in service, logistics and assistive robotics, where a robot could be trained in a digital copy of a new warehouse, hospital or home and operate there reliably after little real-world adaptation. More broadly, they concern any domain in which models are trained on simulated or reconstructed data and deployed on real sensor data, such as autonomous driving or inspection robots.

## 6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / **Outline of the current state of research in the topic of the doctoral dissertation**

*Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).*



Deep learning models for navigation and manipulation need more experience than real robots can provide, so they are usually trained in simulation. Yet performance in simulation can poorly predict real-world performance unless the simulator is carefully tuned [1]. The sim-to-real gap is an instance of learning under distribution shift: domain adaptation theory bounds the error of a model on the target domain by its error on the source domain, the discrepancy between the two distributions, and the joint error of the best single model on both domains [2]. Used as motivation only, the bound suggests three levers: a training distribution that covers reality, a model insensitive to the remaining discrepancy, and, since invariance cannot reduce the joint error, a simulation corrected towards reality.

The first lever is exploited by domain randomization, which varies simulator parameters such as textures so that reality appears as one more variation [3]. Since overly wide randomization leads to conservative behaviour, later works shape its distribution: they fit it to a few real rollouts [4] or maximize its entropy while preserving task success [5]. Yet they act on a few global parameters of a synthetic simulator, not on errors that vary across a scene.

The second lever is invariance. Domain-adversarial training aligns features across domains [6]. In robotics, aligning the joint distributions of observations and actions in simulated and real data improved co-trained policies, i.e., models that map observations to actions, in reality [7]. However, invariance with low source error does not guarantee transfer, as it can increase the joint error term [8], and full invariance may discard task information. Effective sim-and-real co-training aligns the two domains while keeping them distinguishable [9], so domain separability alone does not show where the gap harms the task. Moreover, invariance is usually enforced at a fixed location, the input or the final features.

A recent development is building digital twins. 3D Gaussian Splatting [10] reconstructs photorealistic scenes from ordinary images. In manipulation, twins have been used to train policies from a scan of the target scene [11], to transfer image-based policies to reality without real data [12] and to guide real-world reinforcement learning from a phone capture [13]. Paired simulated and real evaluations showed that scenes visually matched to real images of two robot setups track the real-world performance of the same policies [14]. In navigation, policies have been trained in twins built from a phone capture of a room [15] and from a monocular video of urban scenes [16]. Which simulation factors matter has been studied mainly globally, e.g., randomization, photorealism and physics realism in real-world manipulation [17], or task-relevant fidelity in robotic ultrasound [18]. It remains open, in either task, which reconstruction errors that vary across a scene harm generalization, and how training variation should follow the reconstruction uncertainty of each region.

It is also unclear where inside a model the gap arises. Probing, i.e., reading out information from hidden states with simple classifiers, showed that action fine-tuning degrades the visual representations of robot policies [19], and the best layers to adapt depend on the type of shift [20]. However, the stage at which task information available in simulation stops being recoverable from real inputs has not been located in models trained in twins.

The remaining gap is usually reduced with little real data. Co-training with a small real set yields large gains [21], and active system identification collects the real trajectories most informative about physics [22]. The closest work uses a twin to find failure-prone configurations for real rollouts [13]. However, these methods correct either the model or the simulation; correcting both from the same selected data, guided by the twin's reconstruction uncertainty, has, to our knowledge, not been studied.

Finally, the gap can be decomposed into factors of variation whose difficulty ordering is largely consistent between simulation and reality [23]. However, it is rarely tested whether measurements taken before a method is applied predict, better than simple indicators such as the size of the gap, whether its benefit carries over to a new scene or to another task.



## References

- [1] Kadian, A., et al. (2020). Sim2Real predictivity: Does evaluation in simulation predict real-world performance? RA-L.
- [2] Ben-David, S., et al. (2010). A theory of learning from different domains. Machine Learning.
- [3] Tobin, J., et al. (2017). Domain randomization for transferring deep neural networks from simulation to the real world. IROS.
- [4] Chebotar, Y., et al. (2019). Closing the sim-to-real loop: Adapting simulation randomization with real world experience. ICRA.
- [5] Tiboni, G., et al. (2024). Domain randomization via entropy maximization. ICLR.
- [6] Ganin, Y., et al. (2016). Domain-adversarial training of neural networks. JMLR.
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- [8] Zhao, H., et al. (2019). On learning invariant representations for domain adaptation. ICML.
- [9] Lei, Y., et al. (2026). A Mechanistic Analysis of Sim-and-Real Co-Training in Generative Robot Policies. arXiv:2604.13645.
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- [17] Jin, R., et al. (2026). Grounding Sim-to-Real Generalization in Robotic Manipulation: An Empirical Study with Vision-Language-Action Models. arXiv:2603.22876.
- [18] Qian, Y., et al. (2026). Task-Relevant Feature-Dynamics Fidelity Enables Zero-Shot Sim-to-Real Transfer for Robotic Ultrasound Scanning. arXiv:2608.29516.
- [19] Kachaev, N., et al. (2026). Don't blind your VLA: Aligning visual representations for OOD generalization. AAMAS.
- [20] Lee, Y., et al. (2023). Surgical fine-tuning improves adaptation to distribution shifts. ICLR.
- [21] Maddukuri, A., et al. (2025). Sim-and-real co-training: A simple recipe for vision-based robotic manipulation. RSS.
- [22] Memmel, M., et al. (2024). ASID: Active exploration for system identification in robotic manipulation. ICLR.
- [23] Xie, A., et al. (2024). Decomposing the generalization gap in imitation learning for visual robotic manipulation. ICRA.

## 7. Pytania i hipotezy badawcze / Research questions and hypotheses

*Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).*

The research carried out in the proposed doctoral dissertation is aimed at studying the following research questions, each on two physical AI tasks: robot navigation and robotic manipulation.

1. **Which reconstruction errors of a digital twin, varying across a scene, determine how well navigation and manipulation models trained in it generalize?** Domain adaptation theory, used here as motivation, bounds the real-world error of a model by its error in simulation, the discrepancy between the simulated and real distributions, and a joint error term. A digital twin of





a room or a tabletop reproduces appearance, geometry and lighting only approximately, and its errors differ between regions of a scene. *Hypothesis*: The harm caused by a reconstruction error is predicted better by the change it induces in the representation of a fixed reference model on held-out paired views than by its magnitude in the image or geometry. Separately, randomizing each region in proportion to its reconstruction uncertainty and task relevance yields better generalization than uniform randomization. The research will replace components of the twin with a more accurate reference and measure the effect. If harm follows error magnitude, this attribution will itself be reported, and the third question will rely on reconstruction uncertainty and action sensitivity only.

2. **Where inside a model trained in simulation does the sim-to-real gap arise, and can aligning representations at that stage reduce it?** Models trained in simulation can rely on features absent in reality, such as reconstruction artifacts. Simulated and real inputs remain distinguishable even in models that transfer well, so the gap should be localized by what the model loses about the task, not by whether the domains can be told apart. *Hypothesis*: In both tasks, there is a first stage of the model at which task information that can be decoded from simulated inputs can no longer be decoded from real ones. Aligning simulated and real representations at that stage, from a few paired views, yields better generalization than aligning them at the input, at the final features or everywhere. The research will use probing of hidden states to find this stage before intervening.
3. **Which real data, chosen under a limited budget, most efficiently correct both the simulation and the model?** Real data are expensive, and existing methods use them to correct either the model or the simulation. However, the joint error term cannot be reduced by invariance, only by making the simulation closer to reality. *Hypothesis*: Correcting both the digital twin and the model from the same selected real data reduces the gap more than correcting either one alone. The data should be selected where the twin is predicted to disagree with reality in task-relevant ways. This disagreement can be estimated from the reconstruction uncertainty, the representation distance between simulated and real views, and the sensitivity of the predicted action. Such selection is expected to beat random and failure-driven selection. The research will compare selection rules at equal budgets in both tasks, counting unlabelled real images as real data.
4. **Do the improvements generalize to unseen environments and from one task to the other, and which properties of the shift explain when they do?** A valuable method should not be tied to a single scene or task. Therefore, the developed methods shall be applied with unchanged settings to many held-out scenes of both tasks, and an improvement found in one task shall be tested in the other. *Hypothesis*: The attribution of the gap from the first question and its localization from the second can be measured in a new scene or task before a method is applied. These measurements predict whether the improvement transfers, also between the tasks. They predict it better than simple indicators, such as the raw size of the gap or the image-level discrepancy. The research will test this prediction across many held-out scenes of navigation and manipulation.

## 8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

*Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).*

The proposed PhD dissertation aims to contribute to the development of the scientific discipline by addressing the generalization of deep learning models trained in simulations built from real data, which remains one of the main obstacles to applying machine learning in physical AI. The working assumption of the dissertation is that the sim-to-real gap is caused mainly by a limited set of task-relevant discrepancies, which may be identified among the reconstruction errors of a digital twin, localized inside the model, and reduced with a small amount of well-chosen real data. The methods





are intended to be general, i.e., not tied to a particular robot, scene or task, and are planned to be examined on robot navigation and robotic manipulation.

First, the research is intended to provide a better understanding of which reconstruction errors of a digital twin harm generalization, and a method for varying the twin accordingly instead of uniformly. Second, it aims to develop a way of finding where inside a deep model the sim-to-real gap arises and of reducing it at that stage. Third, it aims to propose a rule for selecting a small amount of real data that corrects both the digital twin and the model. Finally, it will examine when such improvements carry over to unseen environments and between tasks. Such knowledge could complement the currently dominant randomization heuristics with more principled and interpretable approaches.

Since transfer from simulation to reality is an instance of learning under distribution shift, the findings may also be relevant to machine learning in general, in particular to domain adaptation, representation learning and active learning. Apart from purely scientific results, the methods could reduce the cost of deploying learning-based robots in new environments, e.g., in service, logistics and inspection robotics. The source code is planned to be released, which could facilitate further research in the field.

## 9. Planowane metody badawcze / Planned research methods

*Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).*

The research will follow the methods commonly adopted in machine learning and robot learning projects. Since no established theory guarantees generalization from simulation to reality, the dominant part of the work will be empirical. Domain adaptation theory, which relates the error of a model in reality to its error in simulation and to the discrepancy between the two, will serve as motivation, and the properties of the proposed methods will be analysed theoretically whenever possible.

The methods will be examined on two physical AI tasks, robot navigation and robotic manipulation, which share the reconstruction tools. Digital twins will be built from real images with neural scene reconstruction, e.g., 3D Gaussian Splatting, and coupled with simulators for navigation (e.g., Habitat) and manipulation (e.g., ManiSkill3), in which models mapping camera images to actions will be trained or fine-tuned, where possible starting from pretrained models.

Since the research does not rely on an own robot, the evaluation will use public data. For navigation, public datasets of real indoor scenes with independent captures of the same scenes (e.g., ScanNet++) will be used: the twin will be built from one capture, and the other will provide real frames and a dataset-based reference simulation. This reference measures the gap caused by reconstruction fidelity, not differences in actuation and sensors. For manipulation, twins of real tabletop scenes will be built from images of public robot datasets, following the visual matching of SIMPLER [14], and compared with published real-robot results of the same policies; new models will additionally be evaluated in simulations whose parameters are hidden from the learner.

For the first research question, variants of the twin in which one type of reconstruction error is corrected will be examined to attribute the gap to types of error, and randomization guided by reconstruction uncertainty and task relevance will be studied. For the second question, simple probes on the hidden states of models will be used to examine at which stage task information decodable in simulation is lost on real inputs, and alignment at that stage will be compared with alternatives. For the third question, ways of selecting a small amount of real data used to correct both the twin and the model will be examined. For the fourth question, the methods will be applied to held-out scenes of both tasks, and it will be examined whether the measured attribution and localization predict the improvement better than simple predictors.



The evaluation will rely on standard metrics, such as success rate, and on the sim-to-real gap, measured on held-out scenes in both tasks and complemented where possible by open-loop measures on real frames. Methods will be compared with relevant baselines under comparable training effort and real data. Statistical methods will be used to provide reliable outcomes, including repeated runs over seeds and scenes and statistical tests for comparisons, and ablation studies will determine the contribution of individual components. Data for training, model selection and final evaluation will be kept separate. If access allows, selected results will be validated on a real robot in cooperation with a robotics laboratory of the university.

Experiments will be implemented mainly in Python with PyTorch, with source code managed in Git, and run on GPU clusters of the Department, WCSS and PLGrid. Code and data will be released whenever licences allow, following open science standards. The findings will be published at leading conferences from the ministerial list, such as NeurIPS, ICML, ICLR, CVPR and RSS, and presented at smaller venues such as MLinPL.

## 10. Streszczenie popularnonaukowe / Abstract for general public

*Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1).* / *The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).*

### Streszczenie popularnonaukowe

Sztuczna inteligencja coraz częściej wychodzi poza ekrany komputerów i trafia do systemów fizycznych, takich jak roboty poruszające się po budynkach lub przenoszące przedmioty. Modele uczenia głębokiego sterujące takimi robotami potrzebują ogromnej liczby przykładów, których zebranie w prawdziwym świecie jest powolne, kosztowne, a czasem niebezpieczne. Dlatego modele te uczą się zwykle w symulacji, a coraz częściej w cyfrowych bliźniakach, czyli wirtualnych kopiach rzeczywistych miejsc odtworzonych na podstawie ich nagrań. Taka kopia nigdy nie jest jednak idealna. Na przykład szklane drzwi lub odbicie w lustrze mogą zostać odtworzone błędnie i zmylić robota. W efekcie model, który dobrze działa w symulacji, w rzeczywistości często zawodzi, zwłaszcza w miejscach, których wcześniej nie widział. Pozostaje to jedną z głównych przeszkód na drodze do szerszego zastosowania fizycznej sztucznej inteligencji. Celem rozprawy jest poprawa zdolności generalizacji modeli uczenia głębokiego trenowanych w cyfrowych bliźniakach, czyli umiejętności poprawnego działania w warunkach innych niż te, w których je trenowano, a zwłaszcza w rzeczywistości. Badania w pierwszej kolejności pozwolą ustalić, które różnice między symulacją a rzeczywistością faktycznie szkodzą modelowi i na którym etapie przetwarzania w modelu przekształcają się one w błędy. Na tej podstawie zostaną opracowane metody, które nauczą model pomijać nieistotne różnice i więcej ćwiczyć tam, gdzie cyfrowy bliźniak jest najmniej dokładny, a także metody wyboru nielicznych rzeczywistych przykładów, które najlepiej poprawiają zarówno model, jak i cyfrowego bliźniaka. Metody zostaną sprawdzone w dwóch równorzędnych zadaniach, nawigacji robotów oraz manipulacji robotycznej, czyli chwytaniu i przestawianiu przedmiotów na stole, w środowiskach, które nie były wykorzystywane podczas uczenia. Wyniki uzyskane w symulacji zostaną porównane z opublikowanymi wynikami prawdziwych robotów, a jeśli będzie to możliwe, sprawdzone również na prawdziwym robocie. Zbadane zostanie też, czy poprawa uzyskana w jednym zadaniu przenosi się na drugie. Oczekuje się, że rozprawa przyniesie lepsze zrozumienie przyczyn, dla których modele uczone w symulacji zawodzą w rzeczywistości, a także metody, dzięki którym roboty będą mogły szybciej, taniej i bezpieczniej uczyć się pracy w nowych miejscach i przy więcej niż jednym zadaniu.



## Abstract for general public

Artificial intelligence is increasingly moving beyond computer screens into physical systems, such as robots that move around buildings or carry objects. Deep learning models that control such robots need a huge number of examples, which are slow, costly and sometimes unsafe to collect in the real world. Therefore, such models are usually trained in simulation, and increasingly in digital twins, that is, virtual copies of real places rebuilt from recordings of those places. Such a copy is never perfect, however. For example, a glass door or a reflection in a mirror may be rebuilt incorrectly and mislead the robot. As a result, a model that works well in simulation often fails in reality, especially in places it has not seen before. This remains one of the main obstacles to the wider use of physical artificial intelligence. The goal of the dissertation is to improve the generalization of deep learning models trained in digital twins, i.e., their ability to work correctly in conditions different from those they were trained in, and in particular in reality. The research will first establish which differences between simulation and reality actually harm the model, and at which stage of processing inside the model they turn into errors. On this basis, methods will be developed that teach the model to ignore irrelevant differences and to practise more where the digital twin is least accurate, together with methods that choose the few real examples that best correct both the model and the digital twin. The methods will be examined on two equal tasks, robot navigation and robotic manipulation, i.e., grasping and moving objects on a table, in environments that were not used during training. The results obtained in simulation will be compared with published results of real robots and, where possible, also checked on a real robot. It will also be checked whether an improvement obtained in one task carries over to the other. The dissertation is expected to bring a better understanding of why models trained in simulation fail in reality, as well as methods that allow robots to learn to work in new places and on more than one task faster, more cheaply and more safely.

## 11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography

*Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.*

May 2027 (planned submission to the NeurIPS 2027 conference, 200 points on the ministerial list).

## 12. Inne / Other comments

*Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).*



### 13. Określenie planowanej formy współpracy z promotorem / **Planned form of cooperation with the supervisor**

*Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.*

It is planned to have weekly meetings with the supervisor. The meetings are planned to consist of results presentations, discussions, reviews and joint work on papers. The scope of meetings is also supposed to include the evaluation of progress and the planning of further activities. Moreover, it is planned to discuss results, issues and recent advancements in the field with other Ph.D. students weekly, with the supervisor's participation. Day-to-day communication will take place by e-mail and instant messaging.

### 14. Opinia promotora pomocniczego / **Opinion of assistant supervisor**

*Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.*

nie dotyczy / not applicable

.....  
Podpis promotora pomocniczego /  
Signature of assistant supervisor

### 15. Podpisy doktoranta i promotora/promotorów / **Signatures of doctoral student and supervisor(s)**

Wybierz datę/Select date

.....  
Data / Date

.....  
Podpis doktoranta / Signature of doctoral student

.....  
Podpis promotora / Signature of supervisor

.....  
Podpis drugiego promotora / Signature of second supervisor